

## The battle of the 64-bit chips continues

The fight between the market leaders Intel and AMD for supremacy in the 64-bit CPU market rages on with renewed vigour. As of now, there are no 64-bit consumer applications, but AMD has covered for this by its versatility—its 64-bit chip is capable of running 32-bit applications too, something that Intel's Itanium 2 chip (previously code-named McKinley) cannot. This means that it would be easier, not to mention more economical, for firms to make the move over to an AMD 64-bit CPU than Intel's Itanium 2.

Till date AMD was making chips for the consumer market, but it hopes to enter the corporate market with its Ham-

mer processors. The desktop versions of the chip, currently codenamed Clawhammer and slated for release at the end of the year, will be sold under the Athlon name. The multi-processor version of the chip for servers, codenamed Sledgehammer and due in the first half of 2003, will be sold under the name Opteron. AMD has also tied up with Microsoft and the latter will prepare Windows to run on the Hammer chips.

Meanwhile, Linux developers such as SuSE are already offering support for Hammer and the next major public update for Linux will include support for Hammer processors.

Most current 64-bit processors are not compatible with PC applications based on x86 architecture. The Itanium can handle some of those applications but suffers from performance loss—a disadvantage AMD hopes it can exploit.

AMD's 64-bit architecture is an expansion of x86 and theoretically is more versatile than any other 64-bit chip.

Intel, of course, believes its

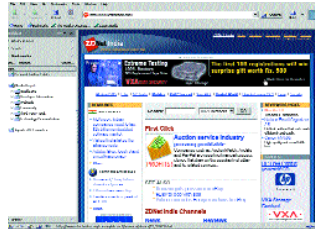
processor is superior in every way. Intel's high-end 64-bit Itanium uses architecture that the company calls IA-64, which builds on existing design technologies and adds some new twists of its own. The Itanium 2 for its part is an improvement over the Itanium—it offers a higher clock speed and other enhancements on the performance front. Intel has also publicly demonstrated that its Itanium 2 processor, which it plans to start shipping at mid-year, will power high-end computer servers and supercomputers up to twice as fast as products from rival Sun Microsystems and its own currently shipping Itanium(1)-based systems.



## Mozilla's finally out!

More than four years in the making, Mozilla 1.0 has been finally released. Netscape Communications, which created the *Mozilla.org* project when it opened up its source code in 1998, explains this delay by saying that they wanted to ensure that the browser stood head to head with other commercial browsers against whom it certainly will be compared.

Gecko, the core browsing engine of Mozilla, supports World Wide Web Consortium recommendations including HTML 4.0, XML 1.0, the Resource Description Framework (RDF), Cascading Style Sheets level 1 (CSS1), and the Document Object Model level 1 (DOM1). Mozilla 1.0 also offers partial support for Cascading Style Sheets level 2 (CSS2), the Document Object



Model level 2 (DOM2), and XHTML. Other standards supported by Gecko include SOAP 1.1, XSLT, XPath 1.0, FIXptr and MathML.

The Mozilla movement was established in 1998 by then-independent Netscape, to create compelling Web-browsing technology to stem the increasing dominance of Microsoft's Internet Explorer. It took more than two-and-a-half years for Netscape to release its first browser using Mozilla technology. Netscape 6 was unanimously criticised as unfinished

and buggy, but future versions of Netscape 6 rectified most of those problems. And with the recent preview release of Netscape 7.0 and AOL's browser shift to Gecko technology, speculations are rife about the prospects of a renewed browser battle with Microsoft, whose Internet Explorer now dominates the Web.

But the biggest effect of the release of Mozilla 1.0 may be beyond the browser. Broadly speaking, Mozilla is a programming tool for building applications that run on almost any operating system. While developers initially concentrated on building a browser, the underlying technology can be used to create many types of applications. Some developers have already branched into making Mozilla instant messaging software, media players and other applications.

### Small is beautiful

California-based Archos Technology has launched the world's smallest and lightest hard disk drive. The Mini HD is a 20 GB hard drive that measures just 4.7x3 inches. It is 0.5 inches thick and weighs just 202.2 grams! Small enough to fit into the palm of your hand, this drive is capable of carrying regular computer files as well as images and digital music. You can simply drag and drop files into or from the Mini HD and it is 'hot swappable'—you do not have to shut down the PC each time you connect or disconnect the device. The drive is multiplatform and you can connect it via USB (USB 2.0 compatible). It is priced at \$200 (approx Rs 10,000).



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